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FACULTY OF ENGINEERING, BUILT ENVIRONMENT AND INFORMATION TECHNOLOGY

BACHELOR IN COMPUTER SCIENCE (HONOURS) (ODL)

BCL2233 COMPUTER ARCHITECTURE

CONTINUOUS ASSESSMENT 30%:
Assignment Topic: Computer Architecture Analysis Based on Professional or Industry Context
DUE DATE SUBMISSION: 25 MARCH 2026, 11:59PM

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By signing below, I acknowledge my understanding and agreement with this declaration.

Student's Signature: Chan Jing Yi Date: 25/3/2026_____

Marking Scheme

Assessment: Architecture Design Analysis Based on Professional Context

Total Marks = 100
Weightage = 30%
CLO measured = CLO3
PLO mapped = PLO2 (Cognitive / Problem Analysis Skills)

Criteria	Marks	Extended Abstract (80–100)	Relational (65–79)	Multistructural (50–64)	Unistructural (40–49)	Prestructural (<40)	CLO	Bloom	Mark Allocation
Professional Context & System Selection	15	Realistic scenario strongly aligned with profession; clear relevance	Appropriate context with good explanation	Context described but limited relevance	Basic or unclear scenario	Poor or missing	CLO3	C3	
Architecture Component Analysis	25	Accurate, integrated explanation of CPU, memory, I/O and data flow	Mostly correct with clear understanding	Several components described separately	Limited explanation	Incorrect concepts	CLO3	C3–C4	
I/O Organisation & Trade-off Analysis	35	Deep comparative analysis with strong justification	Clear analytical comparison	Basic evaluation without depth	Minimal analysis	No meaningful analysis	CLO3	**C4**	
Architecture Recommendation & Justification	25	Logical recommendation strongly supported by analysis	Appropriate recommendation with reasoning	Basic suggestion with limited justification	Weak justification	Missing or unsupported	CLO3	C4	

Overall Feedback

First Marker:

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Total Mark: _____

Signature	Date

Second Marker:

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Total Mark: _____

Signature	Date

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PART A: Professional Context Identification (15 marks)

This report will discuss the retail pharmacy profession, specifically the daily operations of a community pharmacy. In a pharmacy setting, tasks such as providing the right medication to the right patient, inventory management, and counseling the patient occur on a daily basis. In this field, it is extremely important to be accurate to provide the right medication for the right patient and to prevent issues such as running out of inventory and expired products, which may impact the business negatively. This environment is a high-pressure situation. To assist in the daily tasks, this report will discuss the Xilnex Point-of-Sale (POS) and Retail Management System.

Xilnex is a cloud-native and omnichannel solution designed for optimal pharmacy performance. It incorporates front-end retail core functions and pharmacy management functions, including unified inventory management for various outlets, Customer Relationship Management for customer loyalty, and secure payment systems. This system is crucial for the management of real-time inventory accuracy, which is necessary for the management of expired drugs and controlled substances. The offline-first design enables the provision of services to patients even during internet outages before synchronizing the data to the cloud once the internet is restored.

PART B: Architecture Component Analysis (25 marks)

2.1 CPU Role and Functions

All local activities are primarily executed by the Central Processing Unit (CPU) of the Xilnex POS terminal, which is usually an Intel Core i5 or its equivalent in tablet-based installations. In a pharmacy context, the CPU performs complex logic calculations such as the FEFO (First-Expiry, First-Out) logic, where it executes algorithms to automatically select batches of medication based on their expiration dates during the dispensing process. Furthermore, the CPU handles high-level cryptographic operations, including AES-256 encryption for all data transmitted to the cloud, ensuring that patient Personally Identifiable Information (PII) remains secure from potential breaches (Sarcouncil, 2025). Finally, it manages the essential fetch-decode-execute cycle for the POS software, handling multiple threads for simultaneous barcode scanning, user interface updates, and background cloud synchronization.

2.2 Memory and Storage Architecture

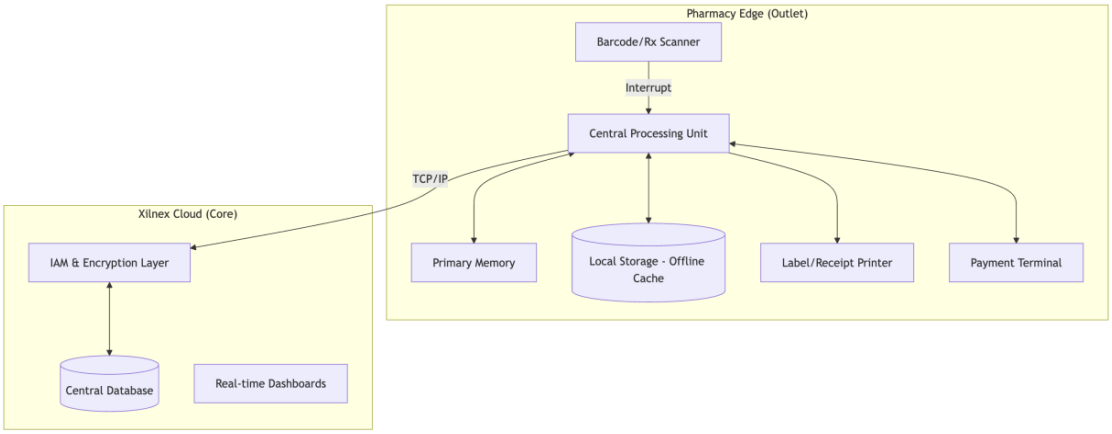
The system uses a multi-tiered memory and storage architecture to strike a compromise between performance and data durability. Primary memory, which consisting of 8GB to 16GB of RAM, is utilized to cache active transaction data and patient records for current sessions, thereby minimizing latency during the checkout process. While for data persistence and resilience, secondary storage in the form of a local Solid State Drive (SSD) stores a local database cache. This allows the

system to function and retain transaction logs locally while the cloud is unavailable, as demonstrated by recent performance assessments (Pagidoju, 2026). The third tier is cloud-based storage, such as Azure or AWS, which serves as the "Single Source of Truth" by permanently storing and backing up all previous sales, global inventory levels, and patient medical histories.

2.3 Input and Output (I/O) Devices

XilNex System uses a variety of input/output devices that are specifically designed for efficient operation in the pharmaceutical industry. The input devices used are a 2D barcode scanner for cataloging products and items in the system, a biometric scanner for the pharmacist, and a keyboard for entering other information as needed, for example, in searching for a product or entering additional information as needed. The output devices used are receipt printers, label printers for prescription information, and display devices for customers. Additionally, bidirectional input/output occurs in the card payment process between the POS system and the EFT payment terminal. In the payment process, the cashier enters the amount of the transaction in the POS system, which then forwards this information to the payment terminal. The customer then inserts his or her card, which then contacts the bank for authorization of the payment transaction. When the bank's decision regarding approval or denial of the payment transaction is sent, this information is then sent from the payment terminal back to the POS system.

2.4 Data Flow and Architecture Diagram



PART C: I/O Organisation & Integration Trade-Off (35 marks)

3.1 Polling vs Interrupt Mechanisms

The Xilnex architecture must balance responsiveness with efficiency by utilizing different I/O strategies. For input devices such as the barcode scanner, an interrupt-driven I/O mechanism is employed. When a barcode is scanned, the device sends a hardware interrupt to the CPU, causing it to pause current background tasks to process the input immediately, which ensures zero-latency feedback for the pharmacist. Conversely, the cloud synchronization process often employs a form of polling or webhooks. The POS system may poll the network status every few seconds to check if a stable connection is available to push locally cached transactions to the cloud. While polling consumes more CPU cycles than interrupts, it is a safer method for managing asynchronous network I/O where the external server's status is unpredictable.

3.2 Real-Time vs Batch Processing

In a retail pharmacy, a hybrid approach to data processing is necessary to maintain operational integrity. Real-time processing is especially important during sales transactions, as inventory must be updated immediately once an item is sold. For instance, when the last bottle of a product is sold, the system should instantly reflect zero stock. If there is a delay, the POS may still show remaining quantity, which can mislead staff into thinking the item is still available when it is actually out of stock. Besides that, it will also cause overselling on e-commerce platforms. On the other hand, batch processing is used for non-critical tasks such as end-of-day financial reconciliation, supplier purchase order generation, and analytical reporting. These tasks are processed in batches during off-peak hours to reduce the load on the CPU and network during busy sales shifts, optimizing overall system performance (Pagidoju, 2026).

3.3 Security vs Usability Trade-Offs

Cloud-based systems like Xilnex face a constant tension between robust security and operational usability. Implementing multi-factor authentication and periodic password resets significantly enhances security but can frustrate pharmacists who require quick access during peak hours. Similarly, end-to-end encryption ensures data integrity but increases CPU overhead and can slow down transaction processing times. However, this trade-off is strictly justified in the pharmacy sector, as the potential cost and ethical implications of a data breach far outweigh the minor inconvenience of a minor login delay (Chavan & Bhoite, 2024).

PART D: Architecture Recommendation (25 marks)

The inherent dependency of current Xilnex deployments on cloud-based validation introduces a critical point of failure during network volatility. To mitigate these risks and enhance operational throughput, the adoption of a Hybrid Edge AI Module integrated directly into the POS hardware is recommended. By employing specialized processing units, such as the Intel Movidius VPU, the system can perform local inference for drug-drug interaction checks and prescription verification. This architectural shift ensures that medication safety protocols remain robust even in the absence of internet connectivity, effectively transitioning the system from a reactive cloud-dependent model to a proactive, resilient edge-computing framework.

Furthermore, the integration of biometric-based Single Sign-On (SSO) protocols addresses the enduring tension between high-level security and retail usability. Rather than relying on traditional, friction-heavy password entry, the use of fingerprint or facial recognition allows for near-instantaneous authentication. This not only improves security by lowering the danger of credential theft, but it also optimizes the pharmacist's workflow, enabling clinical attention to stay on patient consultation. By offloading these computationally intensive validation and encryption tasks to a dedicated co-processor, the primary CPU is liberated from background overhead. The ensuing improvement in responsiveness fulfills the triple goal of dependability, security, and administrative efficiency by guaranteeing a smooth user interface even during times of intense data synchronization.

References

Pagidoju, R.T., 2026. Cost-Performance Analysis of Cloud-Based Retail Point-of-Sale Systems: A Comparative Study of Google Cloud Platform and Microsoft Azure. *arXiv preprint arXiv:2601.00530*.

Sarcouncil Journal of Engineering and Computer Sciences. (2025). *The Role of Cloud POS in Modern Retail Platforms*.

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